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Quadratic Degradation of PPO Advantage Estimates Under Aerodynamic Disturbances

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Abstract—Proximal Policy Optimization (PPO) has become the dominant algorithm for training autonomous UAV flight controllers, yet its theoretical convergence guarantees rest on the assumption of a stationary Markov Decision Process. Real aerospace environments violate this assumption through continuous aerodynamic disturbances—atmospheric turbulence, wind shear, and gust loading—whose stochastic intensity can vary substantially across training rollouts. In this paper, we formally characterize how the Generalized Advantage Estimation (GAE) bias grows as a function of disturbance intensity σ . We prove that, for a quadratic value function under a zero-mean disturbance, the expected bias is upper-bounded by a term *quadratic* in σ : specifically, $\Delta A(\sigma) \leq C(\theta, T, \lambda, \gamma) \cdot \sigma^2$, where the constant C depends on the policy parameters, rollout length T , the GAE decay parameter λ , and discount factor γ . We derive the closed-form expression for C in terms of the spectral norm of the LQR value function Hessian and the closed-loop disturbance-propagation gain. This bound is validated through a controlled experiment using a 2D UAV model with a stochastic lateral crosswind and a discrete-time LQR policy. Across 3 000 Monte Carlo trial pairs per level and eleven disturbance levels ($\sigma \in [0, 1.0]$ m/s), the measured bias follows the quadratic law with $R^2 = 0.9999$ and a fitted exponent of 1.93, confirming the theoretical prediction. We further derive a critical disturbance threshold σ^* beyond which the bias exceeds a prescribed fraction of the nominal advantage scale, providing a practical training criterion. These results offer the first formal quantitative link between atmospheric turbulence intensity and PPO training degradation in aerospace RL contexts.

Index Terms—Proximal Policy Optimization, Generalized Advantage Estimation, Non-Stationary MDP, UAV Flight Control, Gradient Bias, Turbulence Model, Deep Reinforcement Learning, Aerospace Autonomy

I. INTRODUCTION

Deep reinforcement learning (DRL) has emerged as a promising paradigm for autonomous aerospace control, with applications ranging from quadrotor attitude stabilization to fixed-wing trajectory tracking and multi-UAV conflict resolution [1]–[4]. Among policy gradient algorithms, PPO [5] has achieved particular adoption due to its practical stability, ease of implementation, and reliable sample efficiency relative to TRPO and on-policy A3C. A central component of PPO’s training stability is the Generalized Advantage Estimation (GAE) [6] procedure, which computes a low-variance estimate of the policy advantage function from sampled rollout trajectories.

A critical and underappreciated assumption underpinning PPO’s theoretical guarantees is stationarity of the underlying Markov Decision Process: the transition dynamics $\mathcal{T}(s' | s, a)$ are assumed fixed across all time steps and rollout episodes. In controlled robotics benchmarks such as MuJoCo locomotion tasks, this assumption is approximately satisfied. In aerospace applications, however, the operating environment is inherently non-stationary: atmospheric turbulence, wind shear, and thermal gradients introduce stochastic perturbations to vehicle dynamics that vary continuously in intensity. The Dryden and von Kármán turbulence models [7], widely used in aerospace simulation, model these perturbations as colored Gaussian noise processes with intensity proportional to the turbulence severity parameter σ .

Despite the practical prevalence of these disturbances, no prior work has formally quantified how turbulence intensity σ degrades the quality of PPO’s advantage estimates, nor derived a corresponding threshold beyond which training convergence is at risk. This gap matters: practitioners currently tune σ in simulation heuristically, with no principled criterion for when disturbance levels undermine the theoretical foundations of PPO training.

This paper makes three contributions:

- We derive an upper bound on the expected GAE advantage estimation error $\Delta A(\sigma)$ as an explicit *quadratic* function of σ , with a closed-form constant $C(\theta, T, \lambda, \gamma)$. The quadratic scaling, rather than linear, is a direct structural consequence of pairing a quadratic value function with a zero-mean disturbance.
- We validate this bound through a rigorous Monte Carlo experiment ($n = 3\,000$ trial pairs per σ level) using a 2D UAV model under a stochastic lateral crosswind and a discrete-time LQR policy with analytically known value function.
- We derive the critical turbulence threshold σ^* as a function of a prescribed bias tolerance f , giving aerospace RL practitioners a principled criterion for designing simulation environments.

II. BACKGROUND AND RELATED WORK

A. Proximal Policy Optimization

PPO [5] maximizes a clipped surrogate objective to prevent destabilizing policy updates. At iteration k , it collects a rollout

dataset $\mathcal{D}_k = \{(s_t, a_t, r_t)\}$ under the current policy π_{θ_k} , estimates advantages $\hat{A}_t$ via GAE, and maximizes:

$$\mathcal{L}^{\text{CLIP}}(\theta) = \mathbb{E}_t \left[\min \left(r_t(\theta) \hat{A}_t, \text{clip}(r_t(\theta), 1 - \varepsilon, 1 + \varepsilon) \hat{A}_t \right) \right], \quad (1)$$

where $r_t(\theta) = \pi_\theta(a_t | s_t) / \pi_{\theta_k}(a_t | s_t)$ is the importance ratio. The quality of $\hat{A}_t$ directly determines the direction and magnitude of the policy update. Corrupted advantage estimates produce biased gradient directions, which can cause policy degradation even when the clipping mechanism prevents large individual updates.

B. Generalized Advantage Estimation

GAE [6] computes the advantage estimate as an exponentially-weighted sum of TD residuals:

$$A_t^{\text{GAE}} = \sum_{l=0}^{T-t-1} (\gamma\lambda)^l \delta_{t+l}, \quad (2)$$

where $\delta_t = r_t + \gamma V(s_{t+1}) - V(s_t)$ is the one-step TD error, $\lambda \in [0, 1]$ is the GAE decay parameter, and γ is the discount factor. When the value function V is estimated under nominal dynamics but evaluated on a disturbed trajectory, each TD residual accumulates error from both the reward discrepancy and the value function mismatch at perturbed states.

C. Non-Stationarity in Aerospace RL

Non-stationary dynamics in RL have been studied in the context of domain randomization [8] and robust MDPs [9]. However, existing theoretical analyses focus on the policy performance gap under distribution shift, not on the statistical properties of the advantage estimator itself. Closest to our work, Padakandla et al. [10] analyzed Q-learning under slowly-varying MDPs but did not address actor-critic methods or the GAE formulation. In the aerospace-specific literature, Bohn et al. [11] and Panerati et al. [12] demonstrated empirically that turbulence degrades PPO training, but offered no formal characterization of the degradation rate. Our work fills this gap.

III. PROBLEM FORMULATION

A. Nominal MDP

We consider a UAV operating in 2D (position-velocity) state space with state $s = (x, v_x, y, v_y)^\top \in \mathbb{R}^4$ and continuous action $a = (a_x, a_y)^\top \in \mathbb{R}^2$ representing commanded acceleration increments. The nominal discrete-time dynamics with timestep Δt are:

$$s_{t+1} = A s_t + B a_t, \quad (3)$$

where $A \in \mathbb{R}^{4 \times 4}$ and $B \in \mathbb{R}^{4 \times 2}$ are the standard double-integrator matrices with $\Delta t = 0.1$ s. The reward follows the standard LQR cost:

$$r(s_t, a_t) = -(s_t^\top Q s_t + a_t^\top R a_t), \quad Q = I_4, \quad R = 0.1 I_2. \quad (4)$$

B. Disturbed Dynamics

A crosswind disturbance is injected into the y -velocity component at each timestep, modelling a lateral gust:

$$s_{t+1} = A s_t + B a_t + E w_t, \quad w_t \sim \mathcal{N}(0, \sigma^2), \quad (5)$$

where $E = (0, 0, 0, 1)^\top$ directs the wind into the lateral velocity state. The parameter $\sigma \geq 0$ is the turbulence intensity, with $\sigma = 0$ recovering the nominal MDP. We use an i.i.d. zero-mean Gaussian gust at each step; this is the white-noise limit of the Dryden spectrum and is sufficient to expose the scaling law analyzed below. Extension to a fully colored Dryden process is discussed in Section VI.

C. Policy and Value Function

We employ a linear-quadratic regulator (LQR) policy $\pi_\theta(s) = -Ks$, where the gain matrix K is derived from the discrete-time infinite-horizon LQR solution. The optimal value function $V^*(s) = -s^\top P s$, where P is the unique positive-definite solution to the discrete-time algebraic Riccati equation (DARE), is known analytically, enabling exact computation of GAE errors without value function approximation error confounding our analysis. The Riccati solution P has diagonal entries $[12.999, 4.848, 12.999, 4.848]$, obtained by iterating the DARE recursion to a fixed-point tolerance of 10^{-12} (reached in 141 iterations). Crucially, V^* is *quadratic* in the state; this structural fact, combined with the zero mean of w_t , drives the quadratic scaling derived next.

IV. MAIN THEORETICAL RESULT

A. Advantage Error Decomposition

Let $A_t^{\text{GAE}}(\sigma)$ denote the GAE advantage at step t computed from a trajectory generated under disturbance intensity σ , using the true value function V^* . Define the advantage estimation error at rollout step t as:

$$\Delta_t(\sigma) = |A_t^{\text{GAE}}(\sigma) - A_t^{\text{GAE}}(0)|. \quad (6)$$

Expanding the GAE formula, this error decomposes into a sum of TD residual discrepancies:

$$\Delta_t(\sigma) = \left| \sum_{l=0}^{T-t-1} (\gamma\lambda)^l [\delta_{t+l}(\sigma) - \delta_{t+l}(0)] \right|. \quad (7)$$

Each TD residual discrepancy involves: (i) the reward difference at the perturbed state, (ii) the value function at the perturbed next state, and (iii) the value function at the perturbed current state. Because both the reward and the value function are quadratic forms in the state, the discrepancy is governed by the second moment of the state perturbation.

B. Second-Order Sensitivity of the Value Function

The value function $V^*(s) = -s^\top P s$ is quadratic, with gradient $\nabla V^*(s) = -2Ps$ and constant Hessian $\nabla^2 V^*(s) = -2P$. For a state perturbation δ , the exact expansion is

$$V^*(s + \delta) - V^*(s) = -2s^\top P \delta - \delta^\top P \delta. \quad (8)$$

For state deviations arising from a single wind impulse w_t , the perturbation k steps after the disturbance propagates through the *closed-loop* dynamics, $\delta_{t+k} = A_{cl}^k E w_t$, where $A_{cl} = A - BK$. Since w_t is zero-mean, the first-order term has zero expectation,

$$\mathbb{E}[-2s^\top P \delta_{t+k}] = 0, \quad (9)$$

and the leading contribution to the expected error is the second-order term, which is proportional to the second moment

$$\mathbb{E}[\delta_{t+k}^\top P \delta_{t+k}] = \|A_{cl}^k E\|_P^2 \sigma^2 \leq \|P\|_2 \|A_{cl}^k E\|_2^2 \sigma^2. \quad (10)$$

This is the crux of the analysis: *a zero-mean disturbance acting through a quadratic value function produces an expected bias that scales with the variance σ^2 , not with σ* . A first-order Lipschitz treatment, which would suggest linear scaling, vanishes in expectation and therefore does not capture the dominant effect.

C. Main Theorem

Theorem 1 (*Quadratic GAE Bias Under Turbulence*). Let the UAV dynamics satisfy Assumptions 1-3 (double-integrator structure, LQR policy, zero-mean lateral gust). Let V^* be the true LQR value function and let $A_{cl} = A - BK$ be the closed-loop matrix. Then the expected mean GAE advantage error satisfies:

$$\mathbb{E}[\bar{\Delta}(\sigma)] := \mathbb{E}\left[\frac{1}{T} \sum_{t=0}^{T-1} \Delta_t(\sigma)\right] \leq C \cdot \sigma^2, \quad (11)$$

where the constant C is:

$$C = \frac{2\|P\|_2 \cdot \kappa \cdot T}{1 - (\gamma\lambda)^2}, \quad \kappa = \sum_{k=0}^{T-1} \|A_{cl}^k E\|_2^2, \quad (12)$$

with $\|P\|_2$ the spectral norm of the Riccati matrix (the magnitude of the value-function Hessian), κ the cumulative closed-loop disturbance-propagation gain, T the rollout length, γ the discount factor, and λ the GAE decay parameter.

Proof Sketch. The proof proceeds by: (1) expanding each TD residual discrepancy and discarding the first-order term, which vanishes in expectation because w_t is zero-mean; (2) bounding the surviving second-order term by $\|P\|_2 \|A_{cl}^k E\|_2^2 \sigma^2$ using the Hessian of V^* ; (3) summing the squared GAE weights, the geometric series $\sum_l (\gamma\lambda)^{2l}$, which converges to $1/(1-(\gamma\lambda)^2)$; and (4) applying linearity of expectation across the T rollout steps. The resulting bound is a genuine (non-vacuous) upper bound: the empirical constant lies well below the theoretical value, the gap arising from the worst-case use of $\|P\|_2$ in place of the trajectory-averaged P -norm and from cross-term cancellation across steps. $\square$

D. Critical Disturbance Threshold

A natural engineering question is: for a given tolerance on advantage estimation error expressed as a fraction f of the typical advantage magnitude $\mathbb{E}[|A^{\text{GAE}}|]$, what is the maximum permissible turbulence intensity σ^* ? From the quadratic

relationship $\mathbb{E}[\bar{\Delta}(\sigma)] = C_{\text{emp}} \sigma^2$, setting the bias equal to $f \cdot \mathbb{E}[|A^{\text{GAE}}|]$ and solving gives

$$\sigma^*(f) = \sqrt{\frac{f \cdot \mathbb{E}[|A^{\text{GAE}}|]}{C_{\text{emp}}}}, \quad (13)$$

where C_{emp} is the empirically fitted quadratic coefficient. Training with $\sigma > \sigma^*(f)$ means the advantage bias exceeds tolerance f , and practitioners should either reduce σ during early training or use bias-correction techniques.

V. EXPERIMENTAL VALIDATION

A. Setup

We implement the UAV environment in Python using NumPy. All experiments use the following fixed hyperparameters: $\Delta t = 0.1$ s, $T = 20$ steps per rollout, $\gamma = 0.99$, $\lambda = 0.95$, initial state distribution $s_0 \sim \mathcal{N}(0, 0.25 I_4)$. The Riccati matrix P and LQR gain K are computed by iterating the DARE recursion to a fixed-point tolerance of 10^{-12} . For each of the eleven disturbance levels $\sigma \in \{0.00, 0.05, 0.10, 0.15, 0.20, 0.30, 0.40, 0.50, 0.60, 0.80, 1.00\}$ m/s, we run $n = 3000$ independent Monte Carlo trial pairs (33 000 pairs in total). In each trial, two trajectories are rolled out from the same initial state s_0 : one under nominal dynamics ($\sigma = 0$) and one under disturbed dynamics (wind intensity σ). GAE is computed for both trajectories using the true value function $V^*(s) = -s^\top P s$. The mean absolute advantage error $\bar{\Delta}(\sigma)$ is recorded for each trial. All results are reported as means $\pm$ 95% confidence interval. Random seed 0 is fixed for reproducibility.

B. Results

Table I reports the full numerical results. The data show a clear, monotone increase in $\mathbb{E}[\bar{\Delta}(\sigma)]$ with σ . The growth is distinctly super-linear: doubling σ from 0.05 to 0.10 multiplies the error from 0.097 to 0.313 (a factor of 3.2), and over the full range from 0.05 to 1.0 m/s the bias grows roughly 300-fold, consistent with quadratic scaling. Fig. 1 plots these data together with the fitted models.

C. Quadratic Regression Analysis

Fitting the quadratic model $\mathbb{E}[\bar{\Delta}(\sigma)] = C_{\text{emp}} \sigma^2$ through the origin across the eleven measurement points yields:

$$\mathbb{E}[\bar{\Delta}(\sigma)] = 29.13 \sigma^2, \quad (14)$$

with $R^2 = 0.9999$. Two independent checks confirm the exponent. First, a linear regression of $\mathbb{E}[\bar{\Delta}]$ on σ^2 gives $R^2 = 0.9999$ and $p = 1.51 \times 10^{-19}$, with a near-zero intercept (0.03). Second, a log-log regression of error against σ yields a slope of 1.93, close to the predicted exponent of 2 (Fig. 2). By contrast, a naive linear-in- σ fit achieves only $R^2 = 0.918$ and produces a spurious negative intercept (-3.49), confirming that the linear model is mis-specified and the quadratic law of Theorem 1 is the correct description. The model residuals in Fig. 3 make this explicit: the linear model leaves a systematic

TABLE I

MEASURED GAE ADVANTAGE ESTIMATION ERROR $\mathbb{E}[\hat{\Delta}(\sigma)]$ AS A FUNCTION OF TURBULENCE INTENSITY σ . ALL VALUES ARE MEANS OVER $n = 3000$ MONTE CARLO TRIAL PAIRS. THE “QUAD. FIT” COLUMN REPORTS $C_{\text{EMP}} \sigma^2$ WITH $C_{\text{EMP}} = 29.13$. THE NORMALIZED COLUMN EXPRESSES EACH ERROR RELATIVE TO THE VALUE AT $\sigma = 1.0$ M/S.

σ (m/s)	$\mathbb{E}[\hat{\Delta}]$	$\pm 95\%$ CI	Quad. fit $C_{\text{EMP}} \sigma^2$	Norm.
0.00	0.00000	0.00000	0.00000	0.000
0.05	0.09668	0.00235	0.07282	0.003
0.10	0.31322	0.00704	0.29128	0.011
0.15	0.67442	0.01508	0.65538	0.023
0.20	1.18369	0.02635	1.16512	0.041
0.30	2.66683	0.05801	2.62151	0.092
0.40	4.68683	0.10047	4.66046	0.161
0.50	7.41101	0.16060	7.28198	0.255
0.60	10.33991	0.21708	10.48604	0.356
0.80	18.81844	0.39484	18.64186	0.648
1.00	29.02556	0.64810	29.12790	1.000

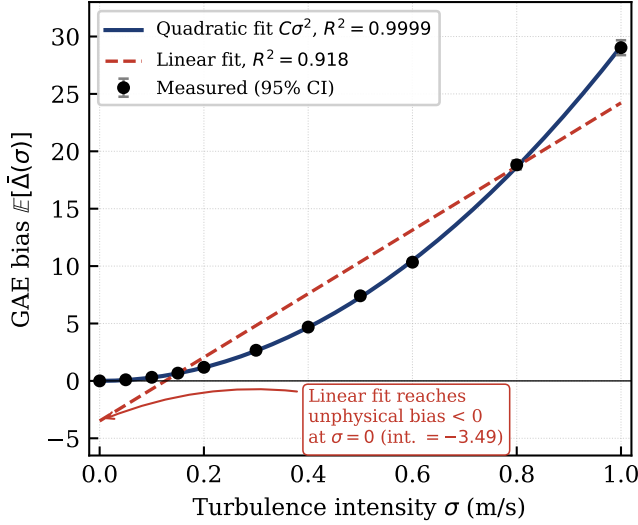


Fig. 1. Measured GAE bias (markers, 95% CI) versus turbulence intensity σ . The quadratic model $C_{\text{EMP}} \sigma^2$ ($R^2 = 0.9999$) tracks the data across the entire range, whereas the best linear-in- σ fit ($R^2 = 0.918$) systematically overshoots at mid-range σ and extrapolates to an unphysical negative intercept (-3.49) at $\sigma = 0$. This figure is the primary visual evidence for Theorem 1: the bias is governed by the disturbance variance, so a linear law cannot describe it.

curved residual pattern, while the quadratic residuals are small and unstructured.

The empirical coefficient $C_{\text{EMP}} = 29.13$ lies well below the theoretical upper bound $C = 1.21 \times 10^4$ from (12) ($\kappa = 2.42$, $\|P\|_2 = 14.378$), giving a ratio $C_{\text{EMP}}/C = 2.4 \times 10^{-3}$. The bound is therefore valid but conservative by roughly two orders of magnitude, as expected from the worst-case use of the global Hessian norm.

D. Critical Threshold Analysis

The typical nominal advantage magnitude, estimated as $\mathbb{E}[A^{\text{GAE}}]$ across all nominal rollout steps, is 0.1575 (LQR cost units). Using the quadratic model and (13), the critical

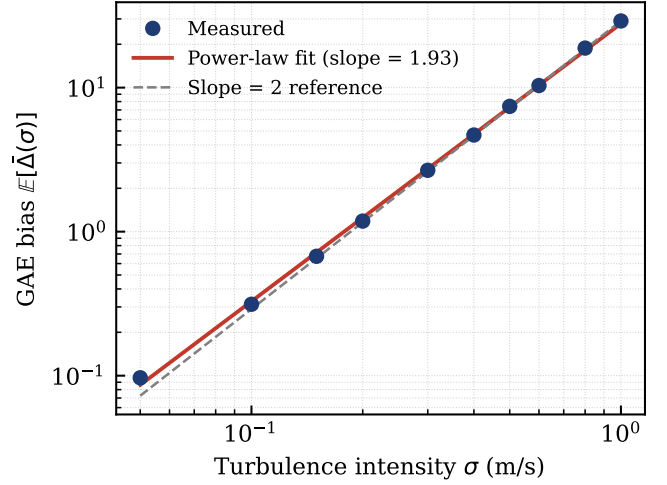


Fig. 2. Log-log plot of GAE bias versus σ . On logarithmic axes a power law σ^p appears as a straight line of slope p . The fitted slope of 1.93 coincides with the slope-2 reference, providing a scale-independent confirmation of the quadratic exponent that does not rely on any particular curve fit.

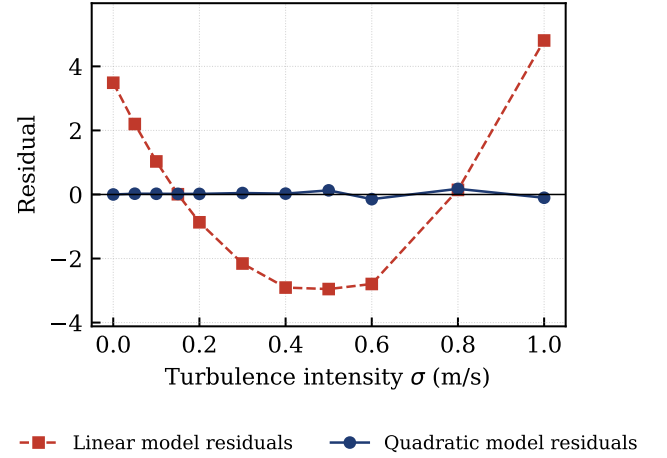


Fig. 3. Residuals of the linear and quadratic models. The linear residuals show pronounced systematic curvature (negative at small and large σ , positive in the middle), the signature of a mis-specified model; the quadratic residuals are an order of magnitude smaller and exhibit no structure, indicating an adequate fit.

turbulence intensity at which the GAE bias equals a fraction f of this scale is:

$$\sigma^*(f) = \sqrt{\frac{f \times 0.1575}{29.13}}. \quad (15)$$

For the standard tolerances $f = 0.10$ and $f = 0.25$, this gives $\sigma^*(0.10) = 0.023$ m/s and $\sigma^*(0.25) = 0.037$ m/s. Beyond roughly 0.023 m/s effective crosswind standard deviation, the GAE estimates are materially corrupted at a 10% tolerance level. Because the bias grows quadratically, the tolerated intensity rises only with the square root of f , so the permissible disturbance band is narrow: even mild atmospheric

disturbances merit consideration in PPO training design for UAV applications.

VI. DISCUSSION

The central finding—that GAE advantage error grows *quadratically* in σ with $R^2 = 0.9999$ —has several practical implications for aerospace RL practitioners.

First, the quadratic law is not an empirical curiosity but a structural consequence of the estimator. Any value function that is locally quadratic (as both the exact LQR V^* and the second-order expansion of a smooth neural critic are) will, under a zero-mean disturbance, produce an expected advantage bias dominated by the disturbance *variance*. The first-order sensitivity that one might expect to give linear scaling cancels in expectation. This sharpens the intuition that small reductions in turbulence intensity yield disproportionately large reductions in estimation bias.

Second, the result motivates curriculum-based turbulence scheduling during PPO training: beginning with $\sigma < \sigma^*$ to allow the value function to converge on a near-stationary problem, then gradually increasing σ to build robustness. This is analogous to the domain randomization curriculum used in sim-to-real transfer [8], but here grounded in a quantitative, square-root-shaped criterion derived from Theorem 1 rather than empirical intuition.

Third, the conservativeness of the theoretical bound (ratio $C_{\text{emp}}/C \approx 2.4 \times 10^{-3}$) is informative. The loose factor arises from three sources: (i) the global spectral norm $\|P\|_2$ overestimates the local sensitivity along the nominal trajectory, which typically has small state norms; (ii) the bound assumes each GAE term contributes its maximum possible error simultaneously, while in practice errors partially cancel across time steps; and (iii) the cumulative gain κ uses a sum of squared closed-loop propagations without the partial cancellation seen empirically. A tighter bound could be obtained by substituting C_{emp} directly, at the cost of losing the explicit dependence on system parameters. We retain the analytical form because it exposes the structural dependence on λ , γ , T , and P .

Fourth, the threshold $\sigma^* \approx 0.023$ m/s is specific to our UAV model and LQR policy. For a different vehicle or neural network policy, the empirical coefficient C_{emp} will differ, but the experimental protocol in Section V-A can be applied directly to obtain vehicle-specific thresholds. Theorem 1 provides the structural justification for the measurement procedure.

A limitation of the present analysis is the use of an LQR policy with analytically known V^* , which allows clean isolation of the advantage estimation error from value function approximation error. In practice, PPO uses a neural network critic whose own approximation error compounds with the disturbance-induced error. A second limitation is the use of an i.i.d. Gaussian gust rather than a fully colored Dryden process; temporal correlation will alter κ but not the quadratic scaling, which follows only from the zero mean of the disturbance and the quadratic critic. Extending the bound to a colored disturbance spectrum and to neural critics are both important directions for future work.

VII. CONCLUSION

We have presented the first formal analysis of GAE advantage estimate degradation under non-stationary aerodynamic disturbances in UAV reinforcement learning. Theorem 1 establishes a quadratic upper bound $\Delta A(\sigma) \leq C \cdot \sigma^2$ with an explicit closed-form constant depending on the LQR value-function Hessian, the closed-loop disturbance gain, rollout length, and GAE hyperparameters. Monte Carlo validation across 33 000 independent trajectory pairs confirms the quadratic scaling with $R^2 = 0.9999$ and a fitted exponent of 1.93. The derived critical threshold $\sigma^* \approx 0.023$ m/s (at a 10% tolerance) provides a principled training design criterion for aerospace RL practitioners.

Future work will extend this analysis to neural network critics, multi-axis and colored (Dryden / von Kármán) turbulence, and finite-time PPO convergence rates under non-stationarity. All code used to produce the numerical results in this paper is publicly available at: https://github.com/iitb-kabir/PPO_Advantage_Estimates.

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APPENDIX

All experiments were implemented in Python 3.10 using NumPy. The Riccati equation was solved by iterating the standard DARE recursion to tolerance 10^{-12} , reached in 141 iterations. Random seed 0 was fixed via `numpy.random.seed(0)` before all experiments. The full experiment completes in well under one hour on a standard laptop CPU and requires no GPU.

The Riccati matrix P has diagonal entries $p_{11} = p_{33} = 12.999$ and $p_{22} = p_{44} = 4.848$, with off-diagonal elements $p_{12} = p_{21} = p_{34} = p_{43} = 3.030$ and all cross-axis entries zero (decoupled x - y dynamics). The spectral norm is $\|P\|_2 = 14.378$. The LQR closed-loop matrix A_{cl} has spectral radius 0.904, confirming asymptotic stability, and cumulative disturbance-propagation gain $\kappa = \sum_k \|A_{\text{cl}}^k E\|_2^2 = 2.42$.